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Lecture due Jan 24, 2024 20:30 IST

When we describe the set of solutions to $\mathbf{Ax} = \mathbf{0}$ as the span of a set of vectors, we want this set of vectors to be as small as possible. In order to eliminate redundant vectors, we need to generalize the notion of linear dependence to 3 or more vectors.

Example 10.1 On the previous page, we saw that $\text{Span} \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right) = \text{Span} \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} \right)$. This is because the vector $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$ is in the span of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. In this case, it may seem obvious that this vector is only adding redundant information because it is a scalar multiple of the other vector. We had a term to describe this, we say that these two vectors are linearly dependent.

What does it mean for 3 or more vectors to be linearly dependent?

Example 10.2 Determine if there are any redundant vectors used to define the subspace

$$\text{Span} \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right).$$

To start, let's see if all vectors are in the span of one vector $\text{Span} \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right)$. If this were the case, we could write

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \text{ as } c \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \text{ for some value } c. \text{ This is not possible. So at most one vector is redundant.}$$

Let's now consider the space $\text{Span} \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right)$. We ask if $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ lies in this subspace. If it does, then it does not add any new information. We see that in fact, it is in this span because

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

This idea of redundence is captured in the definition of what it means for 3 or more vectors to be linearly dependent.

Geometric definition: A collection of vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n$ is **linearly dependent** if there is at least one vector $\mathbf{v}_i$ that lies in the subspace spanned by the other vectors.

Vectors that are not linearly dependent are called **linearly independent**.

Reduction to 2 vectors:

Example 10.3 Let's consider this definition of linear dependence in the case of two vectors: $\mathbf{v}_1, \mathbf{v}_2$. These vectors are linearly dependent if one of these vectors, say $\mathbf{v}_j$ is in the subspace $\text{Span}(\mathbf{v}_i)$. Since $\text{Span}(\mathbf{v}_i) = \{c\mathbf{v}_i, c \text{ any number}\}$, for $\mathbf{v}_j$ to be in this subspace implies that $\mathbf{v}_j = c\mathbf{v}_i$ for some number c . This is equivalent to saying that both vectors lie on the same line through the origin, that is one is a scalar multiple of the other. (This was the definition we gave for linear dependence of two vectors.)

More small examples:

Example 10.4 The vectors $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are linearly dependent because $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ is in the span of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$:

$$\begin{pmatrix} 3 \\ 5 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 5 \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Example 10.5 Are $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, and $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ linearly dependent? The vector $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ is not in the span of $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

and $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$, but we don't know yet if they are linearly independent. They are linearly dependent because $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ is in the span of $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

Testing linear dependence

To have a systematic way of testing for linear dependence, it is helpful to have a definition in terms of an algebraic relationship that we can test.

Equivalent algebraic definition: A collection of vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are **linearly dependent** if there exist scalars $c_1, \dots, c_n$ **not all zero** such that

$$c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n = \mathbf{0}.$$

Proof that the two definitions are equivalent

We show that the two definitions are equivalent. First suppose that there exist scalars, not all zero, so that

$$c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n = \mathbf{0}.$$

Find the first nonzero scalar c_i . Then this equation can be rewritten as

$$\mathbf{v}_i = -\frac{c_1}{c_i} \mathbf{v}_1 - \dots - \frac{c_{i-1}}{c_i} \mathbf{v}_{i-1} - \frac{c_{i+1}}{c_i} \mathbf{v}_{i+1} - \dots - \frac{c_n}{c_i} \mathbf{v}_n,$$

which means that $\mathbf{v}_i$ is in the span of the the other vectors.

Now we show that the logical implication holds in the other direction as well. Note that if one of the vectors, say $\mathbf{v}_n$, is in **Span** $(\mathbf{v}_1, \dots, \mathbf{v}_{n-1})$, then $\mathbf{v}_n$ can be written as a linear combination of the other $\mathbf{v}_i$:

$$\mathbf{v}_n = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_{n-1} \mathbf{v}_{n-1}$$

for constants $c_1, \dots, c_{n-1}$. This is equivalent to saying that there exists a nonzero vector $\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_{n-1} \\ -1 \end{pmatrix}$ such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_{n-1} \mathbf{v}_{n-1} - \mathbf{v}_n = \mathbf{0}.$$

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Remark 10.6 If a collection of $n - 1$ vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n-1}$ are linearly independent, to test if $\mathbf{v}_1, \dots, \mathbf{v}_n$ are dependent, it is enough to check if $\mathbf{v}_n$ is in the span of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n-1}$.

Why? Suppose that the n vectors are linearly dependent. Then there is a nonzero vector $\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_{n-1} \\ c_n \end{pmatrix}$ such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_{n-1} \mathbf{v}_{n-1} + c_n \mathbf{v}_n = \mathbf{0}.$$

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If $\mathbf{c}_n$ were equal to zero, this would imply that

$$\mathbf{c}_1\mathbf{v}_1 + \mathbf{c}_2\mathbf{v}_2 + \cdots + \mathbf{c}_{n-1}\mathbf{v}_{n-1} = \mathbf{0}.$$

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Example in $\mathbb{R}^3$:

Suppose we have 2 linearly independent vectors $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$. These two vectors span a plane inside of $\mathbb{R}^3$.

Now suppose we have a third vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$. This vector is **linearly dependent** if and only if lies in the plane spanned by the first two vectors. This is true if and only if we can write it as a linear combination of the first two vectors.

Linear independence



So those would be independent.
But now, let me stick in a third vector, $\mathbf{v}_3$.
Independent or dependent, now, those three vectors?
So now n is three here.
I'm in two-dimensional space-- whatever-- I'm in the plane.
I have three vectors that I didn't draw so carefully.
I didn't even tell you what exactly they were.
But what's this answer on dependent or independent?
Dependent.
Why?
How do I know those are dependent?

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Algebraic definition of linear independence concept check

1/1 point (graded)

Are the vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ linearly independent?

☒ Yes. They are linearly independent.

☐ No. They are linearly dependent.



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Solution:

Yes. They are linearly independent. Let c_1 , c_2 , and c_3 be numbers such that

$$c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Simplifying, we see that

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

so c_1 , c_2 , and c_3 must all be zero. Hence the three vectors are linearly independent.

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Geometric definition concept check

1/1 point (graded)

Two linearly independent vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in $\mathbb{R}^3$ span the plane defined by the equation $x - 2y + z = 0$. Find

the value c such that $\mathbf{v}_1$, $\mathbf{v}_2$ and $\begin{pmatrix} 3 \\ 1 \\ c \end{pmatrix}$ are linearly dependent.

$c =$

 Answer: -1

Solution:

Because $\mathbf{v}_1$ and $\mathbf{v}_2$ are already linearly independent, it follows that for the three vectors to be linearly dependent, $\begin{pmatrix} 3 \\ 1 \\ c \end{pmatrix}$ must lie on the plane spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$. This happens when $\begin{pmatrix} 3 \\ 1 \\ c \end{pmatrix}$ satisfies the defining equation of the plane

$$3 - 2(1) + c = 0,$$

that is when $c = -1$.

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10. Linear dependence and independence

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